Yaswanth Deevi

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SUMMARY

Machine Learning engineer specializing in deep learning and surrogate-assisted optimization. Experienced in building efficient neural models and hybrid metaheuristic frameworks for large-scale combinatorial problems. Strong focus on model performance, convergence stability, and computational efficiency.

EXPERIENCE

Research Intern Jul 2025 – Sep 2025

Defence Research and Development Organisation (DRDO), Hyderabad

- Designed and implemented a hybrid evolutionary–deep learning framework for the P-Center facility location problem, integrating genetic algorithm search with DNN surrogate models to guide candidate solution evaluation across large-scale graphs.
- Replaced expensive exact fitness computations with a trained feedforward DNN surrogate, reducing per-generation evaluation cost significantly and achieving a **10x improvement in convergence speed** over baseline evolutionary approaches.
- Engineered adaptive Gaussian mutation strategies with dynamic sigma scaling based on population diversity metrics, enabling broader exploration in early generations and precise exploitation near convergence — resulting in a **70% reduction in total computational cost**.
- Collaborated with senior DRDO researchers to document methodology, prepare internal technical reports, and present findings — gaining experience translating research outcomes into actionable engineering recommendations.

TECHNICAL SKILLS

Programming: Python, C++, SQL, Bash scripting
Deep Learning: PyTorch, TensorFlow, Keras, CNNs, RNNs, Transfer Learning, Model Quantization, Model Pruning, ONNX
Machine Learning: Scikit-learn, XGBoost, LightGBM, Gradient Boosting, Random Forests, Logistic Regression, K-Means, PCA, SHAP
Optimization: Evolutionary Algorithms, Genetic Algorithms, Metaheuristics, Surrogate Modeling, Combinatorial Optimization
Data & Analytics: NumPy, Pandas, SciPy, Matplotlib, Seaborn, Statistical Hypothesis Testing
MLOps & Tools: Git, Docker, Linux, AWS, Jupyter, VS Code
Concepts: Computer Vision, Deepfake Detection, Attrition Modeling, Customer Segmentation, Time-Series Analysis

PROJECTS

Anomaly Detection in Building Heating Systems 2025 GitHub Repository

- Developed an FP-Growth association rule mining-based streaming anomaly detection pipeline for multi-room building heating system data; utilized a sliding window approach with window size=6 records, stride=3, and 50% overlap to achieve detection latency ¡ 30s while avoiding boundary effect “missed faults” compared to traditional tumbling window approaches
- Designed average-based discretized transaction generation with outside temperature (OO room) as context feature; utilized frequency-weighted scoring with tightened association rule filters (lift, leverage, conviction) to mitigate false alarms.
- Implemented a pattern stability analysis module that chronologically cross-validates mined rules across N temporal folds, computing per-rule prevalence, confidence drift, and a composite stability score.
- Evaluated the association rule-based pipeline against three unsupervised baselines (Isolation Forest, K-Means, DBSCAN) on identical feature vectors; generated agreement matrices, per-room disagreement matrices, and ROC-style curves to quantify method complementarity.

Deepfake Detection System 2025 GitHub Repository

- Designed a CNN architecture based on Xception for binary video deep fake detection using the Celeb-DF dataset with depth-wise separable convolutions for efficient feature extraction.
- Utilized transfer learning with progressive unfreezing of layers for fine-tuning the top Xception layers to preserve low-level features while adapting to deep fake features.
- Designed a preprocessing pipeline using MTCNN for face detection, alignment, and augmentation (random flip, brightness adjustment, compression simulation) for robust generalization across datasets.
- Assessed model performance using F1 score, ROC AUC, and confusion matrix analysis to identify failure scenarios with varying levels of compression, lighting, and subject variety.

Hybrid Optimization for P-Center Problems 2025 GitHub Repository

- Implemented a surrogate-assisted evolutionary search framework with feedforward DNNs as fitness surrogates replacing expensive distance evaluations in intermediate generations.

- Developed adaptive Gaussian mutation with sigma scaling based on landscape awareness, which is useful for exploration in early generations and exploitation towards convergence.
- Obtained 8- to 10-fold convergence acceleration compared to vanilla genetic algorithms, with solution quality within 1.5% of optimality for standard OR-Lib and TSPLIB instances.
- Carried out ablation studies to compare full exact evaluation, surrogate evaluation, and their combinations; analyzed trade-offs in surrogate evaluation frequency and solution accuracy.

Customer Segmentation and Churn Prediction 2024 GitHub Repository

- Designed 8 behavioral features (tenure, monthly spend, NPS, payment delay, login frequency, contract type) from 2000 synthetic customer records using Standard Scaler for normalization.
- Implemented K-Means clustering with K = 4 clusters using elbow method and PCA-based 2D visualization for customer segmentations with actionable segments: Champions, Loyalists, Dormant Customers, and At-Risk Customers.
- Developed a Gradient Boosting classifier with 300 estimators, learning rate 0.05, and a depth of 4 for predicting customer churn; achieved **CV ROC AUC Score: 0.87 ± 0.01** using 5-fold stratified cross-validation.
- Exported the full pipeline results (cluster profiles, ROC curve, feature importances, confusion matrix, monthly trend of churners) as structured JSON for building an interactive retention strategy dashboard.

HR Analytics and Attrition Modeling 2024 GitHub Repository

- Completed end-to-end analysis on the IBM HR Analytics dataset (1,470 records, 35 features); completed EDA on various features such as demographics, satisfaction levels, tenure, and salaries to understand the attrition phenomenon.
- Used rigorous statistical hypothesis tests such as the Mann-Whitney U Test on continuous variables, the Chi-square Test on categorical variables, and point-biserial correlation on continuous-binary variables; the variables with the greatest significance on attrition were overtime, job satisfaction, and years since last promotion.
- Developed Logistic Regression (Class Balanced, C = 1.0) and Random Forest Classifier (200 estimators, max depth 10) on an 80-20 stratified split; used the Logistic Regression classifier and Random Forest Classifier to understand the decision-making process.
- Retention analysis on 8 cohorts of quarterly hired employees (2022-2023); retention rates at 3 months, 6 months, 12 months, 18 months; monthly hire and exit trends..

EDUCATION

M.S. in Computer Science & Engineering Expected Jul 2027
Blekinge Tekniska Högskola, Sweden
Relevant Coursework: Research Methodologies, Advanced Machine Learning

B.Tech in Computer Science & Engineering Dec 2025
Jawaharlal Nehru Technological University Hyderabad
Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Networks, Object Oriented Programming, Python Programming

CERTIFICATIONS

- IIT KGP AI4ICPS – Advanced Machine Learning & Industrial Cyber-Physical Systems
- Smart Interviews – Data Structures, Algorithms & System Design